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2ºano RSI
Ano-letivo:24/25
UC:SOOS
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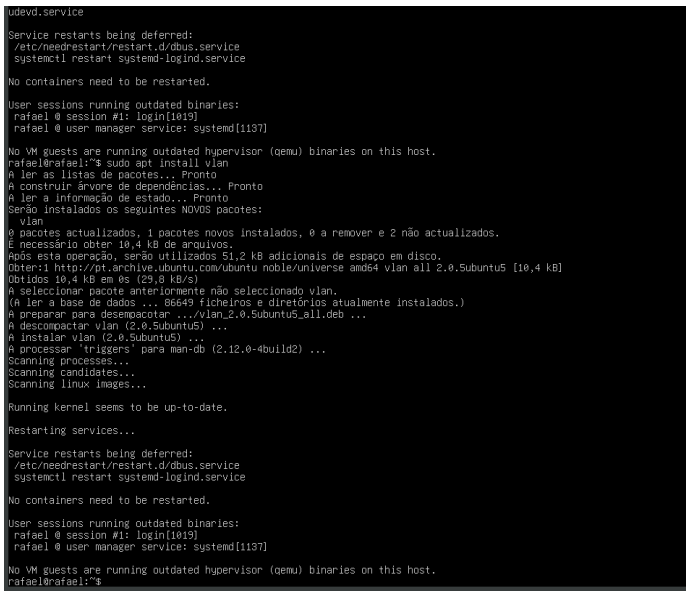
Configuração de VLAN no Ubuntu:

1. Certifique-se de que o pacote `vlan` está instalado:

```bash

sudo apt install vlan
```



```
udevd.service

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart systemd-logind.service

No containers need to be restarted.

User sessions running outdated binaries:
 rafael @ session #1: login[1019]
 rafael @ user manager service: systemd[1137]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
rafael@rafael:~$ sudo apt install vlan
A ler as listas de pacotes... Pronto
A construir árvore de dependências... Pronto
A ler a informação de estado... Pronto
Serão instalados os seguintes NOVOS pacotes:
  vlan
0 pacotes actualizados, 1 pacotes novos instalados, 0 a remover e 2 não actualizados.
É necessário obter 10,4 kB de arquivos.
Após esta operação, serão utilizados 51,2 kB adicionais de espaço em disco.
Obter:1 http://pt.archive.ubuntu.com/ubuntu noble/universe amd64 vlan all 2.0.5ubuntu5 [10,4 kB]
Obtidos 10,4 kB em 0s (29,8 kB/s)
A seleccionar pacote anteriormente não seleccionado vlan.
(A ler a base de dados ... 86649 ficheiros e diretórios atualmente instalados.)
A preparar para desempacotar .../vlan_2.0.5ubuntu5_all.deb ...
A descompactar vlan (2.0.5ubuntu5) ...
A instalar vlan (2.0.5ubuntu5) ...
A processar 'triggers' para man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart systemd-logind.service

No containers need to be restarted.

User sessions running outdated binaries:
 rafael @ session #1: login[1019]
 rafael @ user manager service: systemd[1137]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
rafael@rafael:~$
```

2. Ative o módulo 802.1q:

```bash

sudo modprobe 8021q
```

```
crontab: installing new crontab
rafael@rafael:~$ sudo modprobe 8021q
[sudo] password for rafael:
rafael@rafael:~$ sudo modprobe 8021q
rafael@rafael:~$ _
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
rafael@rafael:~$ sudo modprobe 8021q
rafael@rafael:~$ _
```

3. Configure uma VLAN no arquivo de rede:

```bash

sudo nano /etc/netplan/01-netcfg.yaml

```

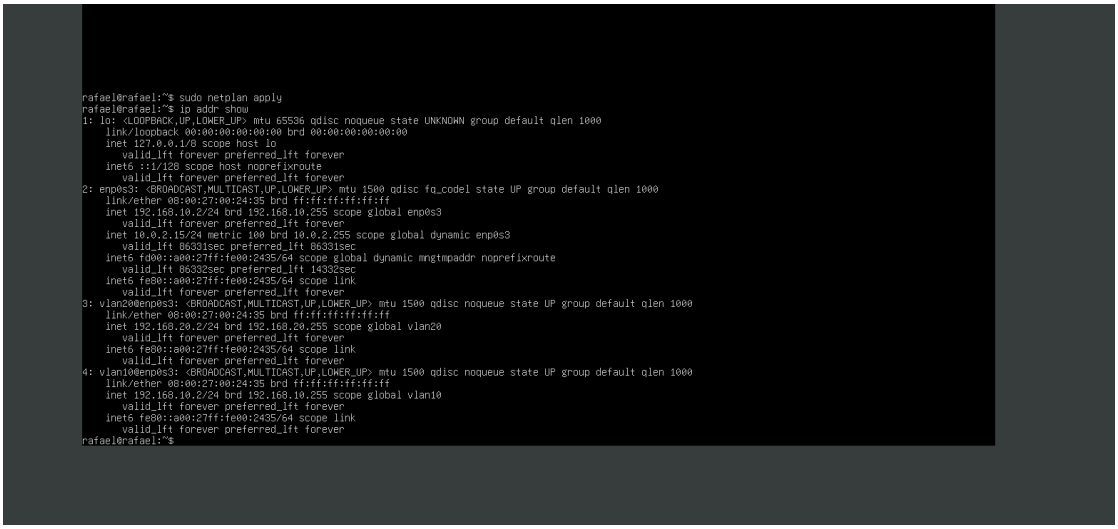
Exemplo de configuração:

```yaml

network:

version: 2

ethernets:

ens33:

dhcp4: no

addresses: [192.168.10.2/24]

gateway4: 192.168.10.1

vlans:

vlan10:

id: 10

link: ens33
```

addresses: [192.168.20.2/24] ```

4. Aplique as configurações:

```bash
sudo netplan apply

```



1.2 Roteamento Dinâmico

Instalação do Quagga ou FRR:

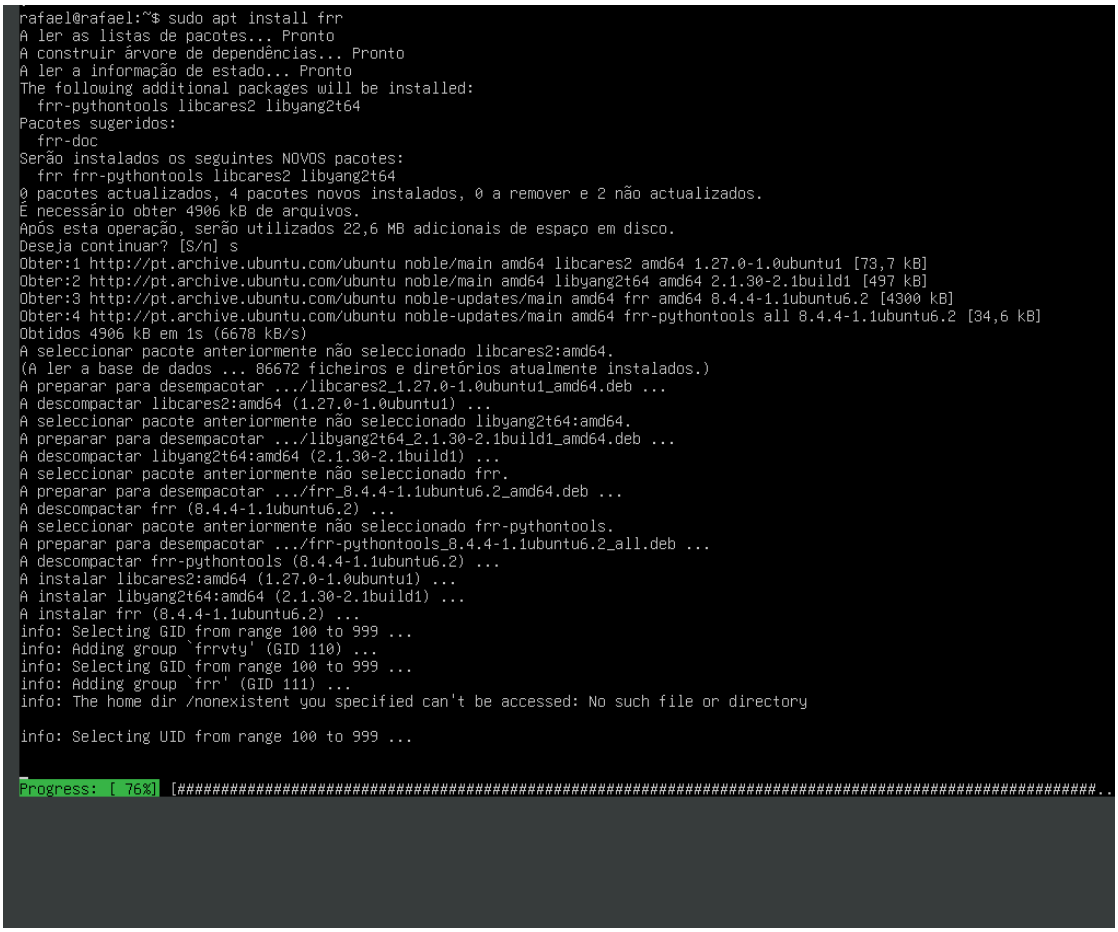
- Para configurar roteamento dinâmico, instale um software de roteamento, como o FRR:

```bash

sudo apt install frr

```

Configuração de OSPF:

1. Edite o arquivo de configuração do FRR:

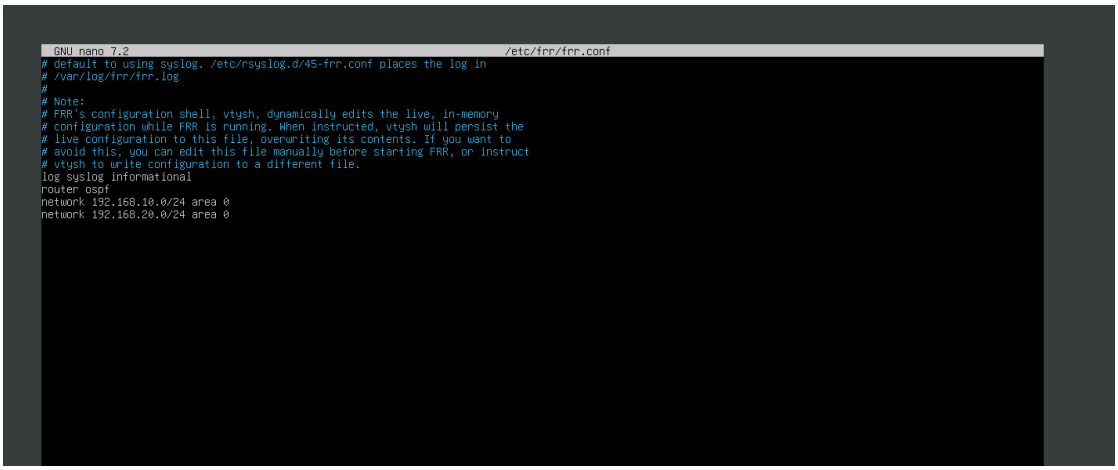
```bash

sudo nano /etc/frr/frr.conf

```



```
rafael@rafael:~$ sudo apt install frr
A ler as listas de pacotes... Pronto
A construir árvore de dependências... Pronto
A ler a informação de estado... Pronto
The following additional packages will be installed:
  frr-pythontools libcares2 libyang2t64
Pacotes sugeridos:
  frr-doc
Serão instalados os seguintes NOVOS pacotes:
  frr frr-pythontools libcares2 libyang2t64
0 pacotes actualizados, 4 pacotes novos instalados, 0 a remover e 2 não actualizados.
É necessário obter 4906 kB de arquivos.
Após esta operação, serão utilizados 22,6 MB adicionais de espaço em disco.
Deseja continuar? [S/n] s
Obter:1 http://pt.archive.ubuntu.com/ubuntu noble/main amd64 libcares2 amd64 1.27.0-1.0ubuntu1 [73,7 kB]
Obter:2 http://pt.archive.ubuntu.com/ubuntu noble/main amd64 libyang2t64 amd64 2.1.30-2.1build1 [497 kB]
Obter:3 http://pt.archive.ubuntu.com/ubuntu noble-updates/main amd64 frr amd64 8.4.4-1.1ubuntu6.2 [4300 kB]
Obter:4 http://pt.archive.ubuntu.com/ubuntu noble-updates/main amd64 frr-pythontools all 8.4.4-1.1ubuntu6.2 [34,6 kB]
Obtidos 4906 kB em 1s (6678 kB/s)
A seleccionar pacote anteriormente não seleccionado libcares2:amd64.
(A ler a base de dados ... 86672 ficheiros e diretórios atualmente instalados.)
A preparar para desempacotar .../libcares2_1.27.0-1.0ubuntu1_amd64.deb ...
A descompactar libcares2:amd64 (1.27.0-1.0ubuntu1) ...
A seleccionar pacote anteriormente não seleccionado libyang2t64:amd64.
A preparar para desempacotar .../libyang2t64_2.1.30-2.1build1_amd64.deb ...
A descompactar libyang2t64:amd64 (2.1.30-2.1build1) ...
A seleccionar pacote anteriormente não seleccionado frr.
A preparar para desempacotar .../frr_8.4.4-1.1ubuntu6.2_amd64.deb ...
A descompactar frr (8.4.4-1.1ubuntu6.2) ...
A seleccionar pacote anteriormente não seleccionado frr-pythontools.
A preparar para desempacotar .../frr-pythontools_8.4.4-1.1ubuntu6.2_all.deb ...
A descompactar frr-pythontools (8.4.4-1.1ubuntu6.2) ...
A instalar libcares2:amd64 (1.27.0-1.0ubuntu1) ...
A instalar libyang2t64:amd64 (2.1.30-2.1build1) ...
A instalar frr (8.4.4-1.1ubuntu6.2) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `frrvty' (GID 110) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `frr' (GID 111) ...
info: The home dir /nonexistent you specified can't be accessed: No such file or directory

info: Selecting UID from range 100 to 999 ...

Progress: [ 76%] [#################################################################################################..
```

2. Adicione a configuração básica do OSPF:

```

router ospf

network 192.168.10.0/24 area 0

network 192.168.20.0/24 area 0

```



3. Reinicie o serviço FRR:

```bash

sudo systemctl restart frr

```

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2. Segurança de Redes

2.1 Configuração de Firewall com UFW

Regras Básicas:

1. Ative o UFW:

```bash

sudo ufw enable

```

```
rafael@rafael:~$ sudo systemctl restart frr
rafael@rafael:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
rafael@rafael:~$ sudo ufw enable
Firewall is active and enabled on system startup
rafael@rafael:~$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)
rafael@rafael:~$ sudo ufw allow from 192.168.10.0/24 to any port 80
Rule added
rafael@rafael:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
80                         ALLOW IN    192.168.10.0/24
22/tcp (v6)                ALLOW IN    Anywhere (v6)

rafael@rafael:~$ _
```

2. Permita apenas conexões SSH:

```bash
sudo ufw allow ssh
```

```
rafael@rafael:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

3. Adicione regras personalizadas:

```bash

sudo ufw allow from 192.168.10.0/24 to any port 80

```

```
rafael@rafael:~$ sudo ufw allow from 192.168.10.0/24 to any port 80
Rule added
```

Verificação de Regras:

- Verifique as regras ativas:

```bash

sudo ufw status verbose

```

```
rafael@rafael:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
80                         ALLOW IN    192.168.10.0/24
22/tcp (v6)                ALLOW IN    Anywhere (v6)

rafael@rafael:~$ _
```

2.2 Monitorização de Tráfego com Wireshark

1. Instale o Wireshark:

```bash

sudo apt install wireshark

```

```
rafael@rafael:~$ sudo apt install tshark
A ler as listas de pacotes... Pronto
A construir árvore de dependências... Pronto
A ler a informação de estado... Pronto
tshark is already the newest version (4.2.2-1.1build3).
0 pacotes actualizados, 0 pacotes novos instalados, 0 a remover e 2 não actualizados.
rafael@rafael:~$
```

2. Capture pacotes na interface desejada:

```bash

sudo wireshark

```

```
rafael@rafael:~$ tshark -D
tshark: Couldn't run dumpcap in child process: Permission denied
1. ciscodump (Cisco remote capture)
2. dpauxmon (DisplayPort AUX channel monitor capture)
3. randpkt (Random packet generator)
4. sdjournal (systemd Journal Export)
5. sshdump (SSH remote capture)
6. udpdump (UDP Listener remote capture)
7. wifidump (Wi-Fi remote capture)
rafael@rafael:~$ 3
3: command not found
rafael@rafael:~$ sudo tshark -i 3
Running as user "root" and group "root". This could be dangerous.
Capturing on 'vlan10'
_
```

3. Analise os pacotes capturados para identificar problemas ou intrusões.

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3. Automação e Scripts

3.1 Introdução ao Bash

Script de Backup:

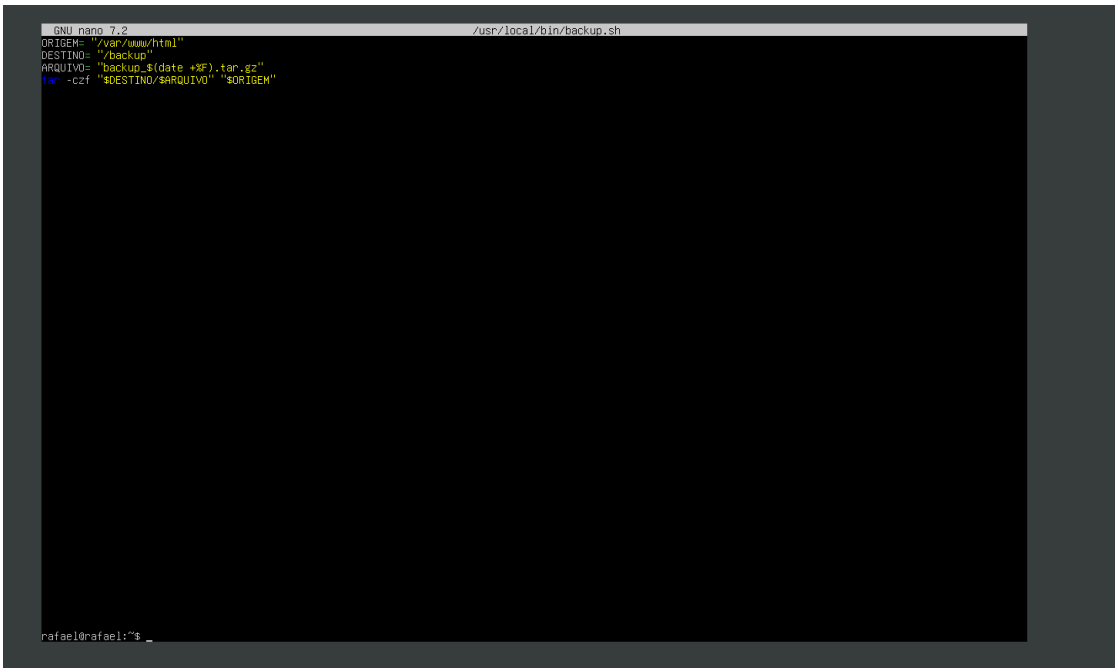
1. Crie um script para backup de arquivos importantes:

```bash

#!/bin/bash

tar -czf /backup/backup_$(date +%F).tar.gz /var/www/html
```

```




2. Torne o script executável:

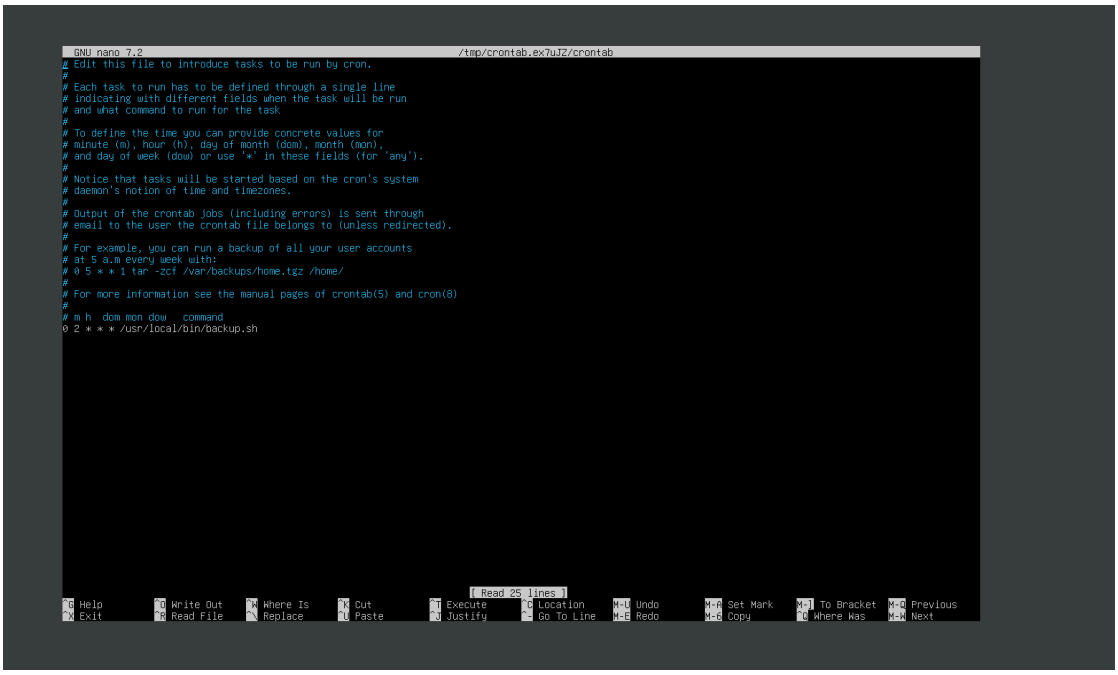
```bash

chmod +x backup.sh

3. Agende o script no cron:
```bash

crontab -e

```

Adicione a linha:

```

0 2 * * * /caminho/para/backup.sh

```



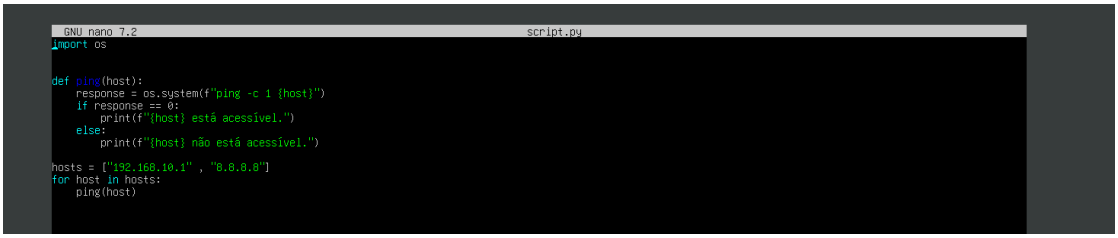
3.2 Introdução ao Python para Redes

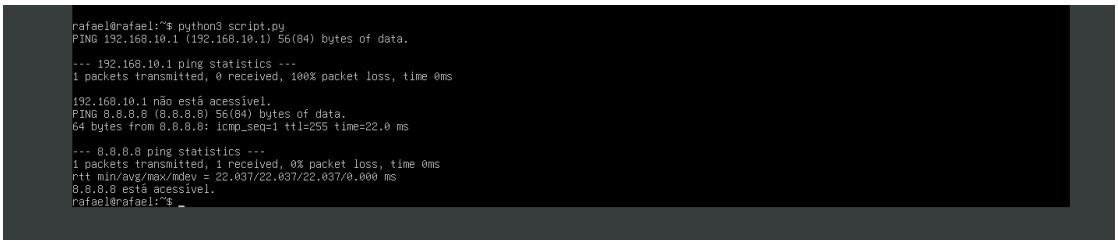
Exemplo de Script para Verificar Conectividade:

1. Crie um script em Python:

```python

import os
```

```
def ping(host):

response = os.system(f"ping -c 1 {host}")  if

response == 0:

print(f"{host} está acessível.")

else:

print(f"{host} não está acessível.")


hosts = ["192.168.10.1", "8.8.8.8"]

for host in hosts:

ping(host)
```



2. Execute o script:
```bash

python3 script.py

```